Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2420)

Key

SECTION – I

1.	D	7.	C	13.	B	19.	C
2.	C	8.	D	14.	C	20.	C
3.	B	9.	C	15.	A	21.	1324
4.	B	10.	A	16.	B	22.	C
5.	C	11.	B	17.	B	23.	B
6.	B	12.	C	18.	2431	24.	2431

SECTION – II

1.	B	6.	A	11.	3	16.	A
2.	10	7.	B	12.	C	17.	B
3.	C	8.	10000	13.	C	18.	D
4.	D	9.	B	14.	D	19.	B
5.	A	10.	4	15.	21	20.	C

SECTION – III

1.	3	7.	90	13.	16	19.	A
2.	C	8.	C	14.	9	20.	A
3.	B	9.	27	15.	B	21.	D
4.	B	10.	A	16.	C	22.	D
5.	D	11.	2	17.	15		
6.	A	12.	50	18.	C		

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 530

1. The first para lays the foundation stone for the whole passage. The resulting entrepreneurial culture has captured the world's imagination and driven the nation to great prosperity. Yet now it is clearly faltering. The second para also states: There's been a secular decline in business formation throughout the country, with a concurrent increase in business dissolution.
Option A: Choice A mentions nothing of the mergers and acquisitions.
Option B: Choice B only indicates what the author acknowledges. The author's attempt, however, is to identify the reasons for the loss. Hence choice B is not the answer.
Option C: Though the author mentions 'consolidation' in paras 3 and 4, he goes on to raise a pertinent question: How does this consolidation impact entrepreneurs? Choice C fails to capture this point. In fact, choice C does not explain the 'entrepreneurial loss' angle at all. Choice C can be eliminated as it does not cover the breadth of the passage.
Option D: Choice D is appropriate as the author's attempt is to identify the reasons for the loss of entrepreneurial spirit. The author places the blame, squarely, on the rise of global corporate behemoths. Choice (D)

2. The second para of the passage mentions Hathaway and Litan's report.
Option A: Choice A is a positive and feel-good statement. It is not the summary of Hathaway's and Litan's report.
Option B: Choice B mentions nothing of the "secular decline in business formation throughout the country". Hence choice B is not the answer.
Option C: Looking at data from all fifty states and all metropolitan areas, Hathaway and Litan conclude ***there's been a secular decline in business formation throughout the country, with a concurrent increase in business dissolution.*** The rate of business formation in

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2019 was almost half of what it was in 1978, with the rate of dissolution somewhat higher than the past couple decades. ...... National response **has not been**, contrary to our myths and history, **one of increased entrepreneurship.** Hence choice C is the apt summary.
Option D: Choice D is again a positive choice and does not capture the main premise of Hathaway and Litan's report.
Only choice C is comprehensive. Choice C is the answer. Choice (C)

3. Option A: Choice A is not true. "Sell to us (buyer)… or get lost" means that there is one buyer. So, monopsony is a market situation in which there is only one buyer (not 'one seller'). Giant firms seek the services of similarly large vendors. New, small entrants into the market will be at pains to form relationships with such firms, and the power imbalance is effectively a monopsony – *sell to us at our price, on our invoice terms, or get lost.*
Option B: "Consolidation of the financial sector" along with "power imbalance" supports choice B. Refer to para 5. New, small entrants into the market will be at pains to form relationships with such firms, and the power imbalance is effectively a monopsony – *sell to us at our price, on our invoice terms, or get lost.* Trying to sell into a world of enormous corporate cartels is considerably more difficult than it was forty years ago. Hence choice B is the answer.
Option C: Consolidation of the financial sector has led to similar dynamics in other industries. .... But choice C does not build on the author's basic argument .... How does this consolidation impact entrepreneurs? Hence choice C is less specific than choice B.
Option D: Choice D gets into details that have not been indicated in the passage, and have no relevance to the key argument made by the author in the passage. Choice (B)

4. Since the question asks about a mistaken assumption that the author corrects, this can only mean assumptions that others may have made, not those in the author's argument (that, thanks to the dominance of big conglomerates, there are very few new ventures, and even those that emerge are hoping not for organic growth on their own but to be swallowed up by the big behind these promising startups are more interested in cashing in on lucrative takeovers (by conglomerates) than on staying in the market and achieving growth themselves. Choice B is the answer.
Option C: Choice C is akin to a feature in the author's argument. It's not an assumption that he overturns.
Option D: There is no discussion on what others believe the financial status of tech start-ups have to be or of what they think the terms of their possible take-overs are like. So this statement is irrelevant. Choice (B)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 522

5. Choice A: Choice (A) ignores the aspect of truth which is highlighted in the penultimate para (The Inquisitors' definition of truth and the process to establish it....) and the last para (....already in possession of the truth, the inquisitors were interested in the rational demonstration of it by each victim). Hence, this is not the answer.
Choice B: In the second sentence of the penultimate paragraph, it has been mentioned that persecution of heresy... hardly seemed a revolutionary step. Hence, Choice (B) is incorrect.
Choice C: Refer to para 3. From the third sentence (to every question there is a right answer), the fourth sentence (The answer is known, but the question must be asked and correctly answered....) and the last sentence (process required that questions be asked in order to produce the correct answer) of the third paragraph, choice (C) is true. Also, it has been given in para 4 that the Inquisitors' definition of truth was arrived at slowly, as was the process which permitted them to establish it. Hence, this is the correct answer.
Choice D: Maintaining notarized manuscripts which became the perpetual record of truth has been mentioned as a detail in the last paragraph but it is not the answer to the question. The purpose was to glorify the methodology and not the outcome. Choice (C)

conglomerates).

Option A: The American entrepreneurial mythos arose in an environment that was perfect for supporting new businesses: rapid growth, technological change, constant competition, limited government intervention. We need to find ways to bring that environment back. This is in line with the author's argument. Hence, choice A is not an assumption that the author corrects.

Option B: The current tech boom might serve as a counter example, (to concentrated industries) but (it doesn't)" means that the mistaken assumption is choice B. This assumption is contested by the author through the point that the entrepreneurs

6. Option A: The idea of establishing order is not a thought that was established during the Inquisition. While it can be argued that prioritizing reason is an element of thought developed during the Inquisition (from "An argument is sometimes made that the assumptions and methods of applied reason where all the key elements of modern intellectual thought can be found were first developed by the Inquisition"), establishing order cannot be a part of the answer.

Option B: The Inquisitors were the first to formalize the idea that to every question there is a right answer. An important point discussed in the third paragraph is that ***the Inquisitors knew the answer and doubt was impossible.*** This point is reiterated in the last para – the Inquisitors were already in possession of the truth. So, choice (B) is true and hence, the answer.

Option C: Choice (C) is negated from the third para – Relativism, humanism, common sense and moral beliefs were all irrelevant to this process because they assume doubt.

Option D: Choice (D) is not the answer to the question as process (which was essential for efficient governance) may have been present even before the Inquisitions. It was not the feature of the Inquisitions only. Choice (B)

7. Option A: From the first sentence of the last paragraph, "Everything the Inquisition did – except the execution of the guilty – took place in secret", it can be inferred that the executions were done openly or in public. Hence, this is true and not the answer.

Option B: Choice (B) can be inferred from the sentences in the last paragraph – And while their power permitted them to do their work on the basis of accusations and denunciations, what they really wanted were complete inquisitions. Being already in possession of the truth, they were interested in the rational demonstration of it by each victim.....to confirm the relationship between a priori truth and assembled fact. Hence, this is true

Option A: This is true as pointed out in the first line of the para. It is ironical that people are finally talking about biodiversity. They are talking about it because biodiversity is getting destroyed. Option A is not the answer.

Option B: A rhetorical question is a question which doesn't expect an answer. Rather, it implies an answer. Similarly, the author's question in the last line of the passage doesn't mean to ask if we would take that gamble. It implies we wouldn't. Hence, Option B is not the answer.

Option C: This is not accurate. True, that it is an analogy that author uses to explain the situation with the biodiversity of earth. Just like we wouldn't take a risk with the life-support system of a patient, we shouldn't take a risk with the health of the planet, looking to increase our revenue. However, the analogy is not to explain that the biodiversity is on the verge of annihilation. The analogy is to explain the importance of biodiversity for the planet not to die. Hence, Option C is the answer.

Option D: This is true. Just like we cannot get rid of a few wires and screws because there are plenty of them, we cannot be reckless towards the biodiversity just because a little loss here and there wouldn't affect the planet. It will. Hence, wires and screws are a metaphor for biodiversity. Option D is not the answer. Choice (C)

and not the answer.
Option C: Choice (C) cannot be inferred. While the passage mentions 'Public silence', we cannot infer that the Inquisitors promoted violence from the passage. Hence, this is the answer.
Option D: Choice (D) can be inferred from the penultimate sentence of the passage – The notary was there to confirm the relationship between a priori truth and assembled fact. Hence, this is true and not the answer. Choice (C)

8. Exactitude means 'preciseness' or 'accuracy'. Hence, choice (A) and choice (B) cannot be the answers. Choice (C) is incorrect because the methodology was glorified and not the outcome. Choice (D) is the correct answer because the exactitude refers to the recording of 'every word of every question and answer' in order to establish the 'relationship between a priori truth and assembled fact'. So choice (D) is correct in the context of the inquisition. Choice (D)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 529

9. All of the options represent various figures of speech. Irony is incongruence between actual result and expected result. Metaphors and similes are comparisons, the latter being direct comparison (Earth is like a patient – example of simile.) Analogy is a simpler situation narrated to explain a bigger. similar, more complicated situation.

10. Option A: Consider this: '*Just as genetic diversity provides species with resilience to environmental change, species diversity increases the resilience of ecosystems.*' From this we can understand that ecosystems that have species diversity (a greater number of species) can adapt better to change. We can understand this from a previous discussion in the passage where the author specifies that more the diversity within a species, more the chance of the species to survive during drastic difficult events. That is because such an ecosystem will have at least some surviving species in case of drastic events, increasing its resilience – the insurance. Option A is therefore apt.
Option B: The author hasn't compared ecosystems with abundant species versus ecosystems with rarer species. In fact, the author clearly states that ecosystems are indeed a mix of both, some common species and a lot of rare species. Hence, Option B is not the answer.
Option C: While ecosystems do have both, the author didn't intend to say ecosystems need these to flourish. Ecosystems need diversity to adapt to future change. That the diversity comes in the form of rare and common species is a natural phenomenon. The author doesn't make it the only condition under which ecosystems flourish. Option C is not the answer.
Option D: Genetic diversity within a particular species makes the species resilient to change. Diversity of various species (not within the same one) makes the ecosystem resilient. So, this option doesn't explain how ecosystem diversity shows its potential to adapt. Option D is not the answer. Choice (A)

11. Option A: While the para states that species interact with each other and with the natural resources, we cannot say that the interaction of species with natural elements is necessary for an ecosystem. So, we cannot really say that the

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 514

ecosystem can only be formed in that one particular way. (Read about Affirmation of the Consequent.) Option A is not the answer.

Option B: This is true. “An ecosystem might be as large as the Great Barrier Reef or as small as the community of sponges, algae and worms hosted on a spider crab’s shell...”. There are big ecosystems and smaller ones, and every ecosystem (including smaller ones) will have different species that interact with each other. Hence, Option B is the answer.

Option C: We cannot quite interpret this from the para. We understand about species dependent on each other and species dependent on natural elements, and all of this creating an ecosystem. We cannot really say if self-dependent species can exist in that ecosystem. Hence, Option C is not the answer.

Option D: While natural conditions like sun, water, soil and air are needed for an ecosystem, the para doesn’t quite specify a ‘set of conditions’ for ecosystems to do well. In fact, the para doesn’t discuss when ecosystems thrive and when they don’t. Hence, Option D is not the answer.

Choice (B)

12. Option A: The passage talks about how genetic diversity improves the resilience of species and species diversity enhances the resilience of an ecosystem. But, it stops short of discussing how diversity in terms of ecosystems can help an environment, although we know that biodiversity helps the environment. So is diversity an important factor here (5 different ecosystems or 5 ecosystems all similar) can help deepen our understanding of the author’s central argument – which is about how important biodiversity is for the environment. Hence, Option A is not the answer.

Option B: While diversity of an ecosystem is important, we don’t know if the number of species is the only thing that matters or if the types of species that exist also matter. Hence, Option B is not the answer.

Option C: Since it is understood that species diversity helps the ecosystem, no further value can be added to the argument by understanding whether species diversity is important. Hence, Option C is the answer.

Option D: Just as in case of Option A, it helps to understand how ecosystems interact with each other and with natural elements. Since health of the environment depends on the health of the ecosystems within, and ecosystems as the passage states depend on each other, this study helps understand the author’s argument. Option D is not the answer.

Choice (C)

13. Option A: This brought home the point to the Nickelodeon creators that there isn’t any diversity in their prime-time characters, since none of them were Latinx. That is one of the things that pushed them towards a Latinx character. Option A is not the answer.

Option B: While this did happen, they didn’t set out with this in mind. The creators wanted someone Latinx because there was an utter lack of diversity. It is not the other way around – that the creators wanted someone who will draw attention and, therefore, brought in a main character different from the rest of the field (all white). Option B is not true. Hence, it is the answer.

Option C: It is indeed mentioned that Pat Buchanan spewing all that hatred motivated them to make a bolder statement for inclusivity. In their own way, the creators were separating themselves from the bigoted talk of Pat Buchanan about getting rid of Spanish speakers. Option C is not the answer.

Option D: This can directly be understood from ‘The creators behind Dora set out to empower Latinx kids and normalize bilingualism.’. The Dora creators and Nickelodeon’s think tank didn’t make any secret of improving representation, after finding that there was no under-18 prime-time character who wasn’t white. Option D is not the answer.

Choice (B)

14. Carlos Cortes’ opinion can be understood from the following lines: “*"Someone came up with the idea we should make her very embedded in one culture – Mexican or Puerto Rican or Cuban or what have you," says Cortes. But they couldn't agree which culture. "I said look, **I think it's important that kids of different Latino backgrounds be able to identify with Dora**.*"” So, Cortes was in favour of a pan-Latino character.

Option A: This option states that a character embedded in one culture will be able to draw kids from only one background. But the idea is to attract kids from all Latino backgrounds. So, the option clearly strengthens Cortes’ argument for a pan-Latino character. Hence, this is not the answer.

Option B: This supports Cortes’ argument that Dora shouldn’t be completely embedded in one particular culture. Pre-schoolers from Latino backgrounds can find commonalities only in that case. Hence, Option B is not the answer.

Option C: This option raises a question mark on Cortes’ argument. Cortes believed that a pan-Latino character would be able to attract all Latino kids. But, if kids cannot relate to the pan-Latino character because of the differences, it contradicts what Cortes thought straightaway, as the attributes of other cultures become a bigger parameter than those particular Latino attributes (common to all cultures) that they can relate to. Option C is the answer.

Option D: This option makes a case for the main character Dora to have a pan-Latino character. The character will then have similarities with kids of various Latino backgrounds. The differences the character will have with specific cultures wouldn't really be of concern to the kids as they take cognizance of similarities more than dissimilarities. Option D is not the answer.

Choice (C)

15. Option A: Consider this: '*We hope that we had an impact on little kids who saw a character who spoke a different language or had a different skin colour than they did,"*.' The Dora creators hoped that they had an impact on little kids. This impact was obviously in not alienating someone different from them (Dora is a Latina) and empathizing with such a character. So, it can be said that her popularity (because she is an icon) will make kids more welcoming of other communities (those who speak a different language and have a different skin colour). Option A is the answer.

Option B: While anti-immigrant rhetoric was taking over the political discourse, the last two paras don't give us any indicators that Dora's storyline is an excellent counter against the rhetoric. We cannot quite infer that it effectively countered the rhetoric although it tried to and that was the hope that it would fight such a rhetoric. Hence, Option B is not the answer.

Option C: This is not true. In fact, the paras imply the contrary, that it is hoped that the little kids did learn a thing or two by observing Dora, a character who spoke a different language and is of a different skin colour. Dora's popularity was a testament to the same. Hence, Option C is not the answer.

Option D: While a film with Dora as its prime character may have political significance, there is nothing in the last two paras that implies that the film is making a powerful political statement and that too against the establishment. Yes, it is trying to convey a message against the anti-immigrant rhetoric, but such an example cannot be used to generalise that films are used to counter the establishment's viewpoint. Option D is not the answer.

Choice (A)

16. Dora's creators hoped that their show will make kids more empathetic towards other communities

to identify with her. This survey will prove them right. Hence, Option C is not the answer.

Option D: This proves that kids who followed Dora's adventures did learn to empathize with those who speak a different language and have a different skin colour. That's why they are less involved in the anti-immigrant rhetoric (read anti-immigrant as anti-Latino as understood from the passage). Option D is not the answer.

Choice (B)

Solution for question 17:

17. Option A: Niall Ferguson's prediction is not central to the author's position and is not accurate either, with respect to what the para is trying to convey. Also, Niall Ferguson wrote about what happened, and didn't 'make a prediction' that takes this option farther away from the essence of the para. Hence, Option A is not the answer.

Option B: This option represents all the attributes of what the author is trying to convey – realists foreseeing an atmosphere of conflict caused by the likelihood of China tweaking rules to suit its own interests, while there is a decline in the powers of America. Hence, Option B rightly depicts the author's position.

Option C: The para doesn't really convey information as presented by the 'realists', who believe that there will be an atmosphere of conflict. It is not explicitly stated that China will go against other countries. Rather, China will use its power to manipulate the system to suit its own interests. Hence, Option C is not the answer.

Option D: The rise of the Chinese hegemony isn't a 'consequence' of the decline of the American era. It is something, the realists believe will happen. Also, this option misses out on an important aspect of what the author states – the reason for the possible conflict between China and other countries. Hence, Option D is not the answer.

Choice (B)

Solution for question 18:

18. Sentence 1 has two parts, and while the first part is independent, the second part is more like a reaction, a conclusion of sorts, as understood from – 'as each day you wiped your memory clear'.

Sentence 2 has a pronoun 'some' referencing a and promote bilingualism.

Option A: Such a survey result will offer validation to Dora's creators as this is something, they set out to do: empower Latinos and promote bilingualism (Spanish being the other language). Hence, Option A is not the answer.

Option B: Such a survey shows that Dora hasn't managed to achieve the purpose for which it was created – to help non-Latino kids like a character that speaks a different language or is of a different colour. Such a survey doesn't validate the intentions of Dora's creators in any way. In fact, it invalidates their aim. Option B is the answer.

Option C: Dora was made pan-Latina to ensure that kids from various Latinx cultures will be able

part of something else – 'Some of the stories'. Which stories is the question.

Sentence 3 has 'them', but it is obvious that the 'them' refers to the stories. Writing the stories, the author had to feel a lot of things until it was all 'gone out of me'. The last part is the clue.

Sentence 4 again references something else with 'they', probably 'stories' again since there is no other plural noun worth considering here.

3 and 4 both depend on 2, but 2 which needs the answer to 'some of what' doesn't have an anchor sentence, or some other sentence it can be derived from. So, 2 comes first. Between 3 and 4, Sentence 3 is about writing them down and

forgetting all about them, whereas 4 is still discussing why they are the stories the author chose to write – he knew those things the best. So, 4 is upstream to 3. So, 243 is one block. 1 is the conclusion – starting with 'because I worked for newspapers' explains the author's habit of forgetting things after he writes them down, as introduced in Sentence 3. So, **2431**. Ans: (2431)

Solutions for questions 19 and 20:

19. Option A: Choice A is not the correct summary as it merely repeats the first and last sentences of the paragraph without focussing on the sentences that provide the central idea of the para. So choice A is not the answer.

Option B: Choice B distorts and extends the argument mentioned. The verbs "bothered" and "concerned" run tangent to the given text. Choice B is also incomplete as a summary.

Option C: Choice C paraphrases the following lines from the paragraph really well: {The value in use of the arts is that they help a society make sense of itself. They generate the symbols and rituals that create a common identity – that is why art and religion are so closely linked.} Choice C best summarizes the para by stressing on the importance of the arts. Society uses art to understand itself. Art helps to create a common identity in a society. The second sentence of choice C also summarizes the first half of the paragraph. Hence choice C is the answer.

Option D: Choice D is incomplete. Choice D is limited to the first two sentences of the para. It does not mention common identity or history.

Choice (C)

Solution for question 21:

21. Sentence 1 is an independent sentence about how more and more rivers are becoming venues for dams.

Sentence 2 is also an independent sentence about the effect of dams. Since it is about the consequences, this sentence is downstream to 1.

Sentence 3 is also a consequence of dams. It also introduces the concept of impeding the flow of nutrients, which precedes the idea in 2 by giving a context to 'increased nutrient retention'.

Sentence 4 talks about 'hotspots for greenhouse gas emissions' which seems to be another disadvantage of dams (in 2 the connection between dams and reservoirs has been made). The connector 'also' in 4 puts this sentence after the issue of 'nutrient retention'.

So, 1324 is the order. Ans: (1324)

Solutions for questions 22 and 23:

22. The statement preceding blank (1) mentions two movies and the statements succeeding blank (1) discuss the two movies. The highlighted sentence talks about deaths, something that is not connected to the statements preceding and succeeding blank (1). Hence, the highlighted sentence does not fit in blank (1).

The statement preceding blank (2) talks about one Pinocchio movie and the statement succeeding blank (2) talks about the other. Both statements logically follow the first sentence of the paragraph which states that there are two Pinocchio films. The highlighted sentence would, therefore, break the flow of the paragraph if inserted in blank (2). Hence, the highlighted sentence does not fit in

20. Option A: This is not the essence of the para, for two reasons. Firstly, human beings are the mass extinction event according to the para. The option states that humans have triggered mass extinctions (plural). Secondly, the para doesn't talk about 'catastrophes.' It calls the mass extinction catastrophic. Option A is not the answer.

Option B: The para doesn't talk about the chances of overturning the mass extinction. It talks about how humans have overturned the food chain, triggering a set of events culminating in mass extinction. Hence, Option B is not the answer.

Option C: This option is apt for mentioning all the main ideas: human beings are the mass extinction event when we look at it from the perspective of planetary history, but this event need not be a tragedy if we remember that nature is in a dynamic state. Option C is the answer.

Option D: The para doesn't mention that overturning the food chain has led to a series of catastrophic events which in turn have led to the mass extinction. The para calls human beings the mass extinction event, which is catastrophic but not tragic. Choice (C)

blank (2).

The statement preceding blank (3) introduces del Toro's movie and the statement succeeding blank (3) elaborates on the death of Pinocchio in del Toro's film. So blank (3) should have a statement that introduces death and connects it to del Toro's movie. The highlighted sentence introduces the deaths of Pinocchio in del Toro's film. The word "plural" in the highlighted sentence is connected to "Pinocchio dies three or four times" in the statement succeeding blank (3). Hence, the highlighted sentence fits in blank (3).

The statement preceding blank (4) already elaborates on Pinocchio's deaths. The highlighted sentence introduces Pinocchio's deaths. Therefore, the highlighted sentence has to come before the statement preceding blank (4). Hence, the highlighted sentence does not fit in blank (4).

Choice (C)

23. The statements preceding and succeeding blank (1) should be consecutive. "effectively loaded the dice" in the statement preceding blank (1) is connected to the "imbalance" in the statement succeeding blank (1). The imbalance is record heat outpacing record cold. The highlighted sentence is not connected to these ideas and would break the flow. Hence, it cannot fit in blank (1).

"The version on this page" in the highlighted sentence has to refer to the "warming stripes" graphics created by Ed Hawkins because in the paragraph given, it is the only mention of something that could resemble a page. The statement preceding blank (2) and the highlighted sentence discuss average temperatures. Hence, the highlighted sentence fits in blank (2).

The rest of the paragraph after blank (2) talks about how every successive decade was hotter than the previous decade. All of these ideas are downstream from the highlighted sentence as they go into further detail regarding what is mentioned in the highlighted sentence. Hence, the highlighted sentence cannot fit in blanks (3) and (4).

Choice (B)

Solution for question 24:

We can understand that 4 follows 2 – 'one small syntactical irritation' elaborated with 'Rather than…'.

We can understand that 1 follows 4 (not necessarily immediately) because it gives another example of what has been mentioned in 4 with a positive connector 'And.'

Now where to place 3? It refers to the previous sentence as a clear attempt. So, if 3 comes after 1, the attempt is what has been depicted in 1, which is just an elaboration of 4. So, 43 is a more logical pair, with 1 adding another example. To confirm that, 1 uses the term 'inner voice' which doesn't sit with 3 – the writer following 'him'. In short, 3 can explain either what is happening in 4 or what is happening in 1. But, if we put 3 in the end, it will look like 4 and 1 are similar cases, but they aren't. Hence, 2431. Ans: (2431)

24. Sentence 1 starts with a positive connector 'And' followed by 'the inner voice' showing that this sentence needs a context.
Sentence 2 is a commentary on Mantel's sentences. It is negative as understood from the word 'caveat'.
Sentence 3 references another sentence ('This') as an attempt to show the writer is behind 'him'.
Sentence 4 talks about a particular habit of writing in the trilogy 'the trilogy has favoured what will doubtless become known as the Mantel Referent Sub-Clause'.

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	1, 2, 10, 11, 13, 15, 20, 22
Difficult	3, 4, 7, 8, 9, 12, 14, 16, 17, 19, 21, 23, 24
Very Difficult	5, 6, 18

SECTION – II

Solutions for questions 1 to 5:

The durations (in minutes) for which the eight persons waited are 145, 135, 110, 85, 50, 25, 10 and 0.
Among the above values, the only value which is twice of another value is 50. From (v), D must have waited for 50 minutes, while the person sitting to the right of D must have waited for 25 minutes.
For (ii), we need two values whose difference is 60. There are three such values (25, 85), (50, 100), (85, 145). Of these three pairs, the second is not possible since D waited for 50 minutes.
Hence, we are left with two possibilities for C and the person sitting to the left of C – (25, 85) and (85, 145).
From (i), A was sitting opposite the person who waited for 110 minutes. Also, A was not the last person to arrive.
From (vi), there is one pair of persons whose waiting time adds up to 110 minutes. The possibilities for this are (110, 0) and (25, 85). The first set of values is not possible as A is sitting opposite the person who waited for 110 minutes, and A did not wait for 0 minutes.
Hence, the person who waited for 25 minutes must be sitting opposite the person who waited for 85 minutes.
If these two persons are sitting opposite each other, then they cannot be C and the person sitting to the left of C.
Hence, C must have waited for 85 minutes and the person sitting to the left of C must have waited for 145 minutes.
Let the positions around the table be represented by the numbers 1 to 8, starting from top in the clockwise direction.
Let D be at 1. The person to the right of D, i.e., the person at 8 must have waited for 25 minutes. The person sitting opposite 8, i.e., the person at 4 must have waited for 85 minutes. Hence, this must be C. The person at 5 must have waited for 145 minutes.

From (vi), there must be two persons who are sitting opposite each other and together waited for more than 240 minutes. Since the person who waited for 145 minutes is sitting opposite the person who waited for 50 minutes (i.e., D), the only possible pair who can wait for 240 minutes or more is (135, 110).

From (i), A must have waited for 135 minutes.

From (iii), F and G can wait for (25, 10) or (10, 0) minutes. In either case, the person who waited for 10 minutes must be F or G.

From (iv), H was not adjacent to F but was adjacent to the person who arrived at 9:25 PM (i.e., the person who waited for 10 minutes). Hence, the person who waited for 10 minutes cannot be F. F must have waited for 25 minutes and G must have waited for 10 minutes.

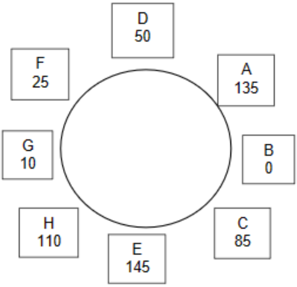
The person at 8 must be F, the person at 7 must be G and the person at 6 must be H.

The only possible places available for A (who waited for 135 minutes) and the person opposite him who waited for 110 minutes are positions 2 and 6. Hence, H must have waited for 110 minutes.

The person at 3 must have been the last person to arrive.

From (i), B cannot be the person who arrived first (since A waited longer). Hence, B must have been the last person to arrive, and E must have been the first person to arrive.

The following diagram provides the arrangement and the time that each person waited:



1. The first person to arrive at the restaurant was E. Choice (B)

2. The person sitting opposite B was G. G waited 10 minutes. Ans: (10)

3. A is sitting opposite H. Choice (C)

4. The persons who arrived after 8:00 PM are C, D, F, G and B. Of these, two pairs of persons (C, F) and (B, G) are sitting opposite each other. Choice (D)

5. Five persons arrived before F. Choice (A)

Solutions for questions 6 to 10:

From the given information, we can get the table below. The table provides the mileage (in '000 km) of each tire at the end of each month. The position of the tires will change every time Pavan changes the tires, which is reflected in the table. Also, FL, FR, BL, BR and S represent the tires at Front Left, Front Right, Back Left, Back Right positions and the spare tire, respectively. The cells highlighted in green indicate that the tire has been discarded and replaced with new tires.

	FL	FR	BL	BR	S
Jan 21	2	2	2	2	0
Feb 21	4	4	4	4	0
	S	FR	BL	BR	FL
Mar 21	4	6	6	6	2
Apr 21	4	8	8	8	4
May 21	4	10	10	10	6
Jun 21	4	2	2	2	8
	BR	FR	BL	S	FL
Jul 21	6	4	4	2	10
Aug 21	8	6	6	2	2
	BR	FR	BL	FL	S
Sep 21	10	8	8	4	2

	BR	FR	BL	FL	S
Oct 21	2	10	10	6	2
Nov 21	4	2	2	8	2
	BR	FR	S	FL	BL
Dec 21	6	4	2	10	4
Jan 22	8	6	2	2	6
	BR	S	FR	FL	BL
Feb 22	10	6	4	4	8
Mar 22	2	6	6	6	10
Apr 22	4	6	8	8	2
	S	BR	FR	FL	BL
May 22	4	8	10	10	4
Jun 22	4	10	2	2	6

6. From the beginning of January 2021 to the beginning of May 2022, Pavan purchased a total of 10 tires. Choice (A)

7. The spare tire had a mileage of 4000 km at the beginning of June 2022. Choice (B)

8. At the beginning of November 2021, the sum of the mileages = 2000 + 0 + 0 + 6000 + 2000 = 10000 km. Ans: (10000)

9. Each tire had a mileage of more than 3000 km only in one month, at the beginning of May 2021. Choice (B)

10. The mileage of the tire in Front Right position more than that of the tire in the Back Right position only in four months, at the beginning of October 2021, March 2022, April 2022 and May 2022. Ans: (4)

Solutions for questions 11 to 15:

Given that none of the traders made a profit greater than ₹40. Hence, the trader who sold the rice at ₹124 must have purchased it for a price greater than ₹84. Therefore, the trader who purchased the rice for ₹85 must have sold it for ₹124, making a profit of ₹39. Let us call this trader the first trader.

The trader who purchased the rice at ₹45, must have sold the rice for less than ₹95. Hence, this trader could have sold it for either ₹78 or ₹84. If this trader sold the rice for ₹84, he will make a profit of ₹39. Since the profits are distinct, this trader must have sold the rice for ₹78, making a profit of ₹33. Let us call this trader the second trader.

From (i), Lalit can be neither the first trader nor the second trader. Lalit could have purchased rice for ₹68 or ₹73 or ₹77. If he purchased the rice for ₹77, he could have made a maximum profit of ₹29 by selling it for ₹106. Hence, he could not have purchased for ₹77. If he purchased the rice for ₹73, he could have made a maximum profit of ₹33. Hence, he could not have purchased it for ₹73. If he purchased it for ₹68, he could have made a profit of ₹38 by selling it for ₹106. He could not have sold at a lower price since the profit in this case will be less than ₹35. Therefore, Lalit purchased the rice at ₹68, sold it at ₹106 and made a profit of ₹38.

From (ii), Manoj must have purchased the rice for ₹73. From (iii), the trader who purchased the rice for ₹77 must have sold it for at least ₹87 to make a profit greater than ₹10. This trader must have sold it for ₹98, ₹106 or ₹124. But he could not have sold it for ₹106 (since Lalit sold it for ₹106) and he could not have sold it for ₹124 (since the first trader sold it for ₹124). Hence, this trader would have sold the rice for ₹98, making a profit of ₹21.

Hence, Manoj would have purchased the rice for ₹73, sold it for ₹84, making a profit of ₹11.

From (iv), neither Piyush nor Omar can be the first trader (since they did not sell the rice for greater than ₹100). Since Omar made higher profit, Omar must have purchased rice for ₹45, sold it for ₹78 and made a profit of ₹33 (i.e., Omar must be the second trader). Piyush must have made a profit of ₹21, by purchasing the rice for ₹77 and selling it for ₹98. Naveen would be the first trader mentioned above.

The following table provides the purchase price, selling price and profit of each trader:

Trader	Purchase Price	Selling Price	Profit
Omar	45	78	33
Lalit	68	106	38
Manoj	73	84	11
Piyush	77	98	21
Naveen	85	124	39

11. Three traders, Omar, Lalit and Naveen, made a profit greater than ₹30 per kg. Ans: (3)

12. Naveen made the highest profit. Choice (C)

13. Statements I and II are true. Choice (C)

14. This question can be answered without solving the entire set. Since the total profit per kg is the difference between the sum of all selling prices and the sum of all purchase prices, the average profit per kg can be arrived at using only the given numbers.
    Sum of all Selling Prices = 490
    Sum of all Purchase Prices = 348
    Average of profits per kg = $\frac{(490-348)}{5} = 28.40$
    Choice (D)

15. Profit made by Piyush = ₹21 per kg. Ans: (21)

Solutions for questions 16 to 20:

From (iv), no student from C4 scored less than 14 in either test. Hence, only A, G, F, K, L and J can belong to C4. Among them, A and G have the same score in GAT, while L and J have the same score in PAT. Hence, at most one from each pair can belong to C4. Since there are 4 students from C4, F and K must belong to C4.

Also, from (iv), every student from C3 scored less than –5 in PAT. From the graph, the only possibilities for C3 are D, M, N and O. Of these, N and O scored the same in PAT and hence, only one of them can be part of C3. Since there are 3 students from C3, D and M must belong to C3.

From (iii), the score of the students from C5 in PAT must be in the bottom 5 (since C1, C2 and C4 together have 10 students). Hence, only D, M, N, O and H can belong to C5. Of these, D and M are from C3. Between N and O, only one can be from C5 (since their scores in PAT are the same). Hence, H and one among N or O must be from C5.

From (ii), the score in PAT of any student from C1 was not less than that of any student from any of C2 or C3 or C4. The score of the students in C1 in PAT must be in the top 4. The possibilities are C, B, L, J. Of these B, L and J have the same score in PAT. Hence, C and one among B, L and J belong to C1.

Of the remaining students, E and I cannot belong to any college but C2.

Since I is from C2, from (ii), the score in GAT of students from C1 must be less than that of I. Hence, C and B belong to C1.

Since E belongs to C2, G cannot belong to C2 (since their scores in PAT are the same). Hence, the only college that G can belong to is C4. The only college that A can belong to is C4. The fourth student from C4 must be L or J.

The following table provides the students from each college:

College	Students
C1	B, C
C2	A, E, I, L/J
C3	D, M, N/O
C4	F, G, K, J/L
C5	H, O/N

16. A belongs to C2. Choice (A)

17. The sum of the scores in PAT of all the students from C2 is 17 (A) + 15 (E) + 16 (I) + 24 (J/L) = 72. Choice (B)

18. N can belong to C3 or C5. Choice (D)

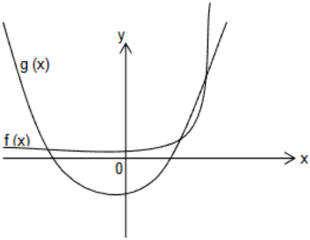
Difficulty level wise summary - Section II

Level of Difficulty	Questions
Very Easy	–
Easy	11, 14
Medium	6, 7, 8, 12, 13, 15
Difficult	1, 2, 3, 4, 5, 9, 10, 16, 17, 18, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. Let $e^x = f(x)$ and $6x^2 - 5x + 6 = g(x)$



$6x^2 + 5x - 6 = 0$

$\Rightarrow (2x + 3)(3x - 2) = 0$

$\Rightarrow x = -\frac{3}{2}$ or $x = \frac{2}{3}$

Let us find the value of $f(x)$ and $g(x)$ at different values of x.

x	-2	0	1	6
$f(x)$	0.14	1	2.7	387
$g(x)$	8	–6	5	240

19. The sum of the scores in PAT of all the students in C4 is 14 (F) + 15 (G) + 16 (K) + 24 (J/L) = 69.
The sum of the scores in GAT of all the students in C4 is 28 (F) + 30 (G) + 25 (K) + 23 (J)/18 (L) = 101 or 106.
The required difference is either 32 or 37.
NOTE: Although Choice (A) does satisfy, the question asks which option BEST satisfies the given condition. Hence, the answer will be choice (B). Choice (B)

20. The sum of the scores in PAT of the students from C1 = 24 + 29 = 53
The sum of the scores in GAT of the students from C5 = 28 + 24/27 = 52 or 55.
From the given condition, the sum of scores in GAT of the students from C5 must be 52. Hence, N should belong to C5 and O should belong to C3. Choice (C)

At $x = -2$ $f(x) < g(x)$
At $x = 0$ $f(x) > g(x)$

Therefore there is one point of intersection between $x = -2$ and $x = 0$
Similarly at $x = 1$ $f(x) < g(x)$
Therefore there is another point of intersection between $x = 0$ and $x = 1$
Similarly at $x = 6$ $f(x) > g(x)$
$\therefore$ There is a third point of intersection between $x = 1$ and $x = 6$.

From the graph, we can conclude, there will not be any other point of intersection of $f(x)$ and $g(x)$ when $x > 6$ (since the exponential curve increased more steeply than the quadratic). Thus there are exactly three points of intersection. Ans: (3)

2. Let the present ages (in years) of Ram, Sita and Bala be r, s and b respectively.
Ram was as old as Sita was $(r - s)$ years ago.

Ages of Sita and Bala then were $[s - (r - s)]$ years and $[b - (r - s)]$ years respectively.
$s - (r - s) = 2\,[b - (r - s)]$
$2s - r = 2\,[b - r + s]$
$r = 2b$
Also $r + b = 120$
$\therefore b = 40$ Choice (C)

3. We compare the squares of the terms as whichever term is greater, its square will also be greater.
$a^2 = 25 + 2\sqrt{136}$
$b^2 = 25 + 2\sqrt{154}$
$c^2 = 25 + 2\sqrt{144}$
$d^2 = 25 + 2\sqrt{156}$

Thus, the second item was sold at cost price, i.e., no profit no loss. Choice (D)

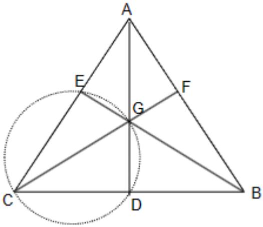
6. It is given that $m > 4$, $m \in z$. $n = m(m - 1)$
$\Rightarrow$ n is even, since the product of any two consecutive numbers is always even.
Now, $n^3 - 6n^2 + 8n$
$= n(n^2 - 6n + 8)$
$= n(n - 2)\,(n - 4)$ = product of three consecutive even numbers. The product of three consecutive even numbers will always have at least one multiple of 4 and exactly one multiple of 6. The third number is of course, definitely even, i.e., a multiple of 2.
$\therefore$ The number $n^3 - 6n^2 + 8n$ will always be divisible by $2 \times 4 \times 6 = 48$. Choice (A)

Now, $\sqrt{156} > \sqrt{154} > \sqrt{144} > \sqrt{136}$

$\therefore d^2 > b^2 > c^2 > a^2$

$\therefore d > b > c > a$ Choice (B)

4.



Consider the cyclic quadrilateral EGDC, AC and AD are secants from the same point to the circle.

$\Rightarrow$ AE. AC = AG. AD

AE = EC = x, say.

And AG = 2GD = 2y, say.

(Since E is mid point of AC, while G is centroid and AD is the median).

Hence $x.\ 2x = 2y.\ 3y \Rightarrow x = \sqrt{3}\ y \Rightarrow$ AE = $\sqrt{3}$ GD

$\Rightarrow$ I is true.

Similarly, if Q_1, Q_2 and Q_3 are all concyclic, then AE = $\sqrt{3}$ GD $\Rightarrow$ AD = $\frac{\sqrt{3}}{2}$ AC and AD = $\frac{\sqrt{3}}{2}$ AB

$\Rightarrow$ AB = AC. Similarly AC = BC can be proved. (i.e., ABC is equilateral). Hence, III is true.

$\Rightarrow$ Only I and III are true. Choice (B)

5. Let the selling price of each item be x

As the overall profit is 20%

$(CP_1 + CP_2)\ 1.2 = 2x$

$\therefore CP_1 + CP_2 = \frac{2x}{1.2} = \frac{5}{3}x$

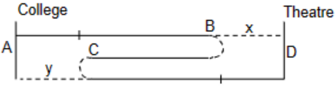
As the first item was sold at 50% profit,

$1.5CP_1 = x$

$CP_1 = \frac{x}{1.5} = \frac{2}{3}x$

$\therefore CP_2 = \frac{5}{3}x - \frac{2}{3}x = x$

7.



Let the distance covered by each of Arif and Chetan on foot be x and y respectively.

In the time Chetan covered y, Bill with five times his speed must have covered $5y$.

$\therefore$ AB + 2BC = $5y \Rightarrow$ BC = $2y$

Similarly for Arif, BD = x and BC = $2x \Rightarrow x = y$

Now, total distance from A to D = $4x$ = 22.5 km

$\therefore$ The bike covered AB + BC + CD = $3x + 2x + 3x$

= $8x$

= 45 km.

To cover 45 km, it took $\frac{45}{30}$ = 1.5 hrs or 90 minutes.

Ans: (90)

8. Let there be x lines in P1 and y lines in P2. The number of points of intersection will be equal to xy.

Given xy = 18 and x < y.

The possibilities are x = 3, y = 6 and x = 2, y = 9

Case I: x = 3, y = 6

The number of parallelograms that can be formed will be equal to ${}^3C_2 \times {}^6C_2 = 45$

Case II: x = 2, y = 9

The number of parallelograms that can be formed will be equal to ${}^2C_2 \times {}^9C_2 = 36$

Therefore, the number of parallelograms formed by P1 and P2 will either be 45 or 36, but definitely not 54. Choice (C)

9. $\sin^2\theta - 2\sin^3\theta = (\sin\theta)(\sin\theta)(1 - 2\sin\theta)$

now, $(\sin\theta) + (\sin\theta) + (1 - 2\sin\theta) = 1$ (i.e., a constant)

$\therefore$ the product will be maximum when

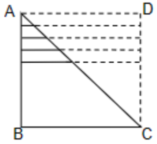
$\sin\theta = \sin\theta = 1 - 2\sin\theta \Rightarrow \sin\theta = \frac{1}{3}$

Therefore, the maximum value of the product is

$\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(1 - \frac{2}{3}\right) = \frac{1}{27} = \frac{1}{A}$ (given)

$\therefore$ A = 27 Ans: (27)

10.



Constructing triangle ADC as shown above we get a square (∵ AB = BC)

Now, the length of each line is equal to the side of the square which is $6\sqrt{2}$

So, the total length of the 12 lines will be $12(6\sqrt{2})$.

But we need only half of this i.e., $6 \times 6\sqrt{2} = 36\sqrt{2}$ cm. Choice (A)

11. Given, $\log_2 (a \log_3 b) = 1$

$\Rightarrow \log_3 b^a = 2$

$\Rightarrow b^a = 9$

Since a and b are both natural numbers, the only possibilities for this are when b = 3, a = 2 and when b = 9, a = 1.

Therefore, the number of ordered pairs (a, b) that satisfy the given equation are 2. Ans: (2)

12. Given that

$2^{100} - 2^{99} - 2^{98} - 2^{97}\ \ldots\ldots\ldots - 2^{50} = k.2^n$

where $\{k, n\} \in N$

Consider $2^n - 2^{n-1} = 2.2^{n-1} - 2^{n-1} = 2^{n-1}$

$\Rightarrow 2^{99} - 2^{98} - 2^{97}\ \ldots\ldots\ldots\ldots - 2^{50} = k.2^n$

$\Rightarrow 2^{98} - 2^{97}\ \ldots\ldots\ldots\ldots - 2^{50} = k.2^n$

.

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$2^{51} - 2^{50} = k.2^n$

$\Rightarrow 2^{50} = k.2^n$

Here k can take any value from 2^0 to 2^{49}.

∴ The greatest value of n is 50 (when $k = 1$)

Ans: (50)

13. Let the first term and the common ratio of the geometric progression be a and r.

Sum of the first 12 terms = $\frac{[a(r^{12}-1)]}{(r-1)}$

Sum of the next 6 terms will be = Sum of the first 18 terms – Sum of the first 12 terms.

$= \frac{[a(r^{18}-1)]}{(r-1)} - \frac{[a(r^{12}-1)]}{(r-1)}$

$= \frac{[a(r^{18}-r^{12})]}{(r-1)}$

Given that $\frac{[a(r^{12}-1)]}{(r-1)} = \frac{6[a(r^{18}-r^{12})]}{(r-1)}$

$r^{12} - 1 = 6(r^{18} - r^{12})$

$(r^6 - 1)(r^6 + 1) = 6r^{12}(r^6 - 1)$

$(r^6 + 1) = 6r^{12}$

$6r^{12} - r^6 - 1 = 0$

Solving for r^6, we get $r^6 = \frac{1}{2}$ or $\frac{-1}{3}$.

However, since $r^6 > 0$, $r^6 \neq \frac{-1}{3}$

∴ Seventh term of the series is $ar^6 = 32 \times \frac{1}{2} = 16$.

Ans: (16)

14. Given expression is $[(999)^{999}]^{999}$. As the units digit is affected only by the last digit of 999, consider $[(9)^{999}]^{999}$.

We know that the units digit of $(9)^n = 9$, if n is odd, and 1, if n is even.

$\therefore [(9)^{999}]^{999}$

$= 9^{999 \times 999}$

$= 9^{(\text{odd number})}$

∴ The units digit will be 9. Ans: (9)

15. Spirit in 1 litre mix of A = $\frac{7}{10}$ litre.

Spirit in 1 litre mix of B = $\frac{4}{13}$ litre.

Spirit in 1 litre mix of C = $\frac{5}{13}$ litre.

Let the required ratio be x : 1

$\frac{\left(\frac{7}{10} \times x + \frac{4}{13} \times 1\right)}{(x+1)} = \frac{5}{13}$

$x = \frac{10}{41}$

Therefore, the required ratio is 10 : 41

Choice (B)

16. Let the actual salary of Nitish be ₹100. For 50% hike his new salary should be ₹150. As the computer operator made a mistake, his salary according to the computer operator is ₹180. Nitish got ₹90 more than the actual salary for the 3 months. His salary has to be reduced by ₹90 from ₹150.

∴ He receives $\left(\frac{60}{150}\right) \times 100\%$ = 40% of his new salary in the fourth month. Choice (C)

17. Let the initial number of coins with Ajay be N

Firstly, N must be even, (since he gave one more than half of N to Bulu). Further, let $\frac{1}{3}\left[\frac{N}{2} - 1\right] = k$, _____ (1) where k is a natural number and

$\frac{2}{3}\left[\frac{N}{2} - 1\right] \geq 2$ ___ (2)

(Since he gave two coins more than one-third of the remaining to Chintu).

Hence, Chintu got a total of $(k + 2)$ coins.

If k is a natural number, from (1), we get N must be of the form $6k + 2$, which also is even (the first requirement).

Further, from (2), N ≥ 8.

Hence, N can be any number of the form $6k + 2$, where

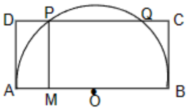
$k = 1, 2, 3\ldots\ldots$

Since it is mentioned that N is a two-digit number, $10 \leq N \leq 99$.

Hence, the possible values of k are 2, 3, 4, ……. 16, i.e., N can assume a total of 16 – 2 + 1 = 15 distinct values. Ans: (15)

18. Given P and Q are natural numbers, the units digit in 5^P is always 5, irrespective of P.
Since Q is less than 7, in order to satisfy the given condition, the units digit in P^Q can be 0 or 1. Further, Since P is a natural number less than 4, the units digit in P^Q cannot be 0. Therefore, the units digit in P^Q is 1. Therefore, the sum of the units digits in P^Q and 5^P, i.e. Q = 6, i.e., the units digit of $P^Q = 1$.
This is possible only when P = 1 or P = 3.
If P = 1, the sum of the units digits 1^Q and 5^1 will be 6, i.e. Q = 6. This is consistent.
If P = 3, for the units digit in 3^Q to be 1, Q should be equal to 4. But Q = 6. This is not consistent.
Therefore, P = 1 and Q = 6 is the only possibility and Q^P i.e. $6^1 = 6$ is the only possible value for Q^P.
Choice (C)

19.



Given AM + MP = 9 → (1)
And MP + BM = 18 → (2)
Let AM = x and PM = kx
$\therefore$ BM = k^2x (In ΔAPB, rightangled at P, AM, PM and BM are always in geometric progression, because $\frac{AM}{PM} = \frac{PM}{BM}$)
(1) $\Rightarrow x(1 + k) = 9$
(2) $\Rightarrow xk(1 + k) = 18$
$\therefore k = 2$ and $x = 3$
$\therefore$ AB = AM + MB = $x + k^2x = 3(1 + 4) = 15$
$\Rightarrow$ Area of semicircle = $\frac{\pi}{2}(7.5)^2 = 28.125\pi$

Alternative solution:

Let the length and breadth of the rectangle ABCD be ℓ and b respectively and let O be the centre of AB. Now, OP = OA $\frac{\ell}{2}$ (Radius of the semicircle).

Let AM be x.
$2x + 2b = 18$ → (1) and
$2(\ell - x) + 2b = 36$ → (2)

Again $OP^2 = OM^2 + PM^2 \Rightarrow \left(\frac{\ell}{2}\right)^2 = (\ell - x)^2 + b^2$
$\Rightarrow \ell x = x^2 + b^2$
From (1), $x^2 + b^2 + 2bx = 81$
$\therefore 81 - 2bx = \ell x$ or
$x(\ell + 2b) = 81$
From (1) and (2), we get $\ell + 2b = 27$,
So $x = 3$, $b = 6$ and $\ell = 15$

Area of the semicircle = $\frac{\pi}{2}(7.5)^2 = 28.125\,\pi$

Choice (A)

20. f(1) = f(0) – f(2) = 3
f(3) = f(2) – f(4) = f(2) – 1
f(2) = f(1) – f(3) = f(1) – f(2) + 1
2f(2) = 4. Hence f(2) = 2
f(3) = f(2) – 1 = 1
Now f(4) = f(3) – f(5). From this, we can say that the value of f(5) = 0
f(4) = 1 = f(3) – [f(4) – f(6)] = 1 – 1 + f(5) – f(7)
From this, f(7) = –1
Choice (A)

21. The quadratic equation whose roots are half the roots of the equation $f(x) = 0$ is $f(2x) = 0$.
The required equation is
$(2x)^2 + 5(2x) + 3 = 0$
$\Rightarrow 4x^2 + 10x + 3 = 0$
Choice (D)

22. $m = \frac{n+76}{n+4} = \frac{n+4+72}{n+4} = 1 + \frac{72}{n+4}$

For m to be an integer, 72 must be divisible by $n + 4$ i.e., $n + 4$ must be a factor of 72.
$72 = 2^3 \times 3^2$ number of factors = (3 + 1) (2 + 1)
= 12
Therefore $n + 4$ can take 12 values of which 1, 2, 3 and 4 are not admissible as n must be positive.
Thus, eight values are possible.
Choice (D)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	–
Easy	3, 5, 11, 16, 21
Medium	2, 6, 8, 10, 13, 14, 15, 17, 19, 20
Difficult	1, 4, 7, 9, 12, 18, 22
Very Difficult	–

Done
Tip: Scroll slowly for the best results